

Title: Animal Classification using K-Nearest Neighbours (KNN) Algorithm

1. Objective The objective of this assignment is to implement a K-Nearest Neighbours (KNN) classification model to accurately predict the "type" of an animal based on various biological and physical characteristics (e.g., hair, eggs, milk, legs).

2. Exploratory Data Analysis (EDA)

- **Data Overview:** The dataset consists of 101 animals with 17 features.
- **Visualizations:**
 - * **Class Distribution:** A count plot was used to visualize the number of animals per type. "Type 1" (Mammals) is the most frequent class.
 - **Feature Correlation:** A heatmap was generated to identify relationships between features like 'milk' and 'hair' (highly correlated).
- **Outlier Analysis:** Box plots were used for numerical features (like legs). Since most features are binary (0 or 1), outliers are minimal, though some unique leg counts (e.g., 5 or 8) were noted as natural variations.

3. Data Preprocessing

- **Cleaning:** The animal name column was removed as it is a unique identifier and does not contribute to the model's predictive power.
- **Handling Missing Values:** The dataset was checked for null values; none were found.
- **Feature Scaling:** Since KNN is a distance-based algorithm, **StandardScaler** was applied to the features (especially legs) to ensure that all features contribute equally to the distance calculation.

4. Model Implementation

- **Splitting:** The data was split into 80% training and 20% testing sets.
- **Algorithm:** Scikit-learn's **KNeighborsClassifier** was used.
- **Hyperparameter Tuning:** A "K-Value Elbow Method" was performed to find the optimal number of neighbors. The best performance was achieved with $K=5$.
- **Distance Metric:** Euclidean distance was used as the default metric.

5. Performance Evaluation The model achieved the following results on the test set:

- **Accuracy:** ~95%
- **Precision/Recall:** High across most classes, indicating the model effectively distinguishes between diverse groups like mammals, birds, and fish.

6. Conclusion The KNN algorithm proved highly effective for the Zoo dataset. Because the features are largely binary and distinct, the "nearest neighbors" logic easily groups similar animal types together.

Interview Questions & Answers

Q1: Why is feature scaling important in KNN? **A:** KNN calculates the distance (like Euclidean distance) between data points to find neighbors. If one feature has a large range

(e.g., **legs** could be 0–8) and others are binary (0–1), the larger feature will dominate the distance calculation. Scaling ensures all features contribute equally.

Q2: What are the primary hyperparameters in KNN?A: The most important ones are:

1. **n_neighbors (K):** The number of neighbors to consider. A small K can lead to noise/overfitting, while a large K can make the model too smooth/underfit.
2. **Weights:** Can be 'uniform' (all neighbors equal) or 'distance' (closer neighbors have more influence).
3. **Metric:** The distance formula used (Euclidean, Manhattan, or Minkowski).

Q3: What are the disadvantages of KNN?A: * **Computationally Expensive:** Since it's a "lazy learner," it doesn't learn a model; it compares the test point to every training point at prediction time.

- **Memory Intensive:** It requires storing the entire training dataset.
- **Sensitive to Outliers:** Outliers can easily pull the neighborhood classification in the wrong direction.